



The Effect of Transferring on Player Performance: An Analysis on NCAA Division I Men's Basketball

An Le, Bates College

Kendall Lucas, Duquesne University

External Advisor: Katerina Wu, University of Pittsburgh

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Motivation

- With new NCAA transferring and eligibility rules, the number of athletes entering the transfer portal has increased in recent years
- Transferring has a huge impact on coaches, players, and programs as a whole

→ **Main question:** Does transferring improve a player's athletic

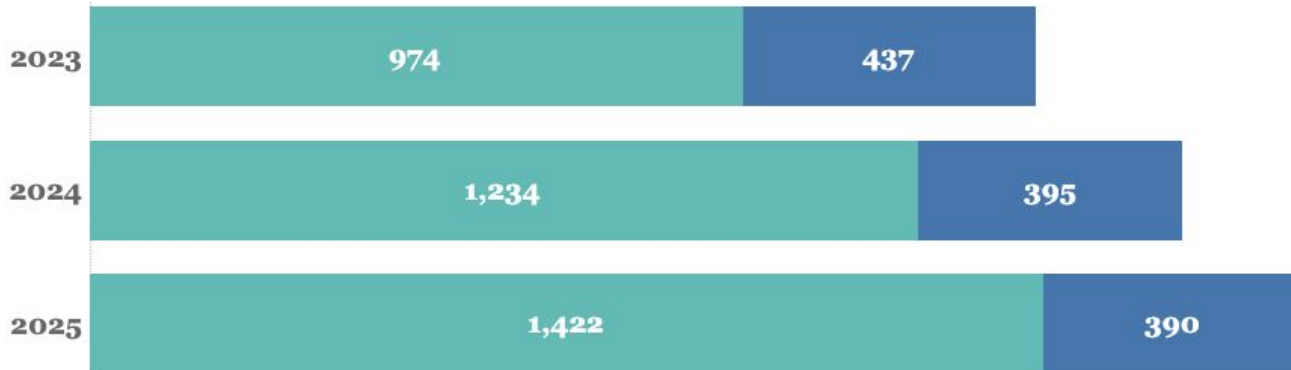
performance?

Sport
Transfer Portal

How many Division I Transfer Portal entrants transferred to another NCAA school?

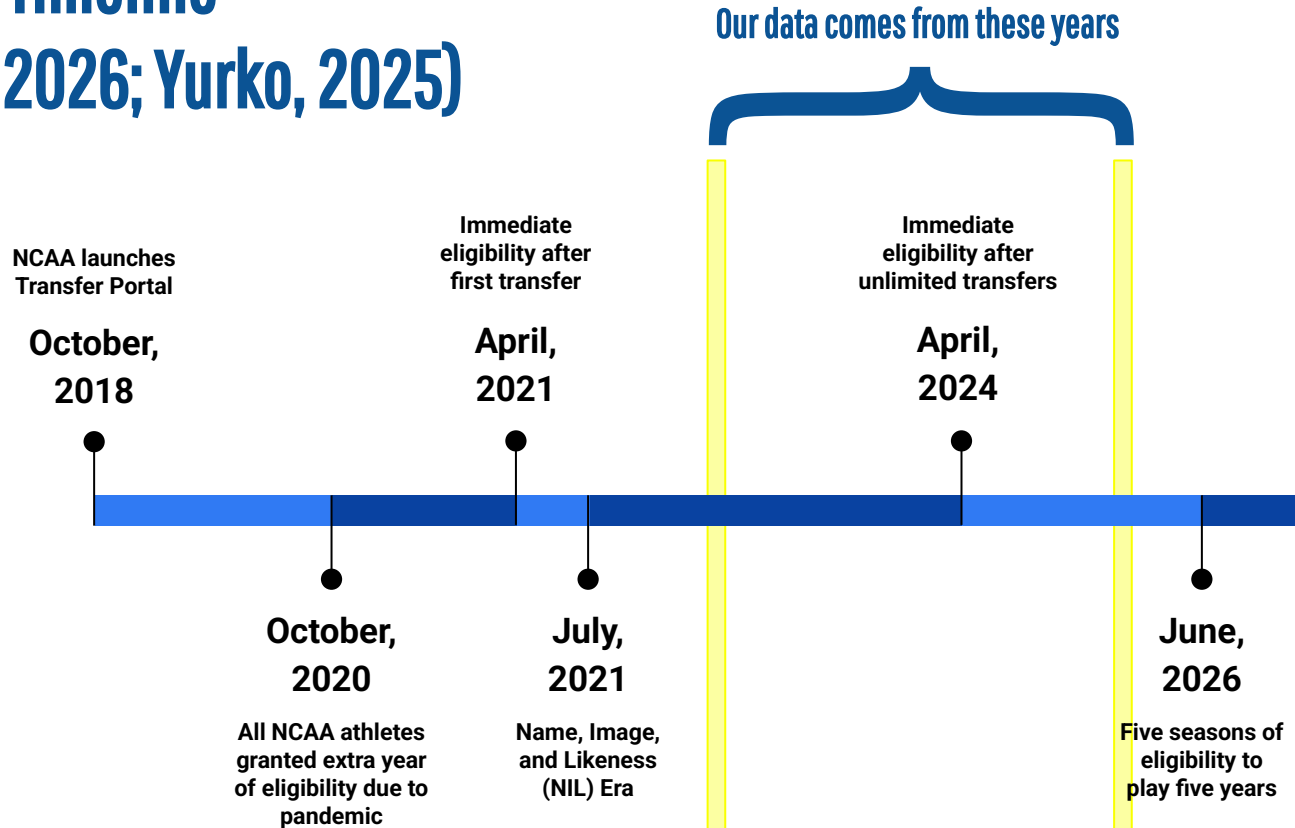
Undergraduate Entrants

Graduate Entrants



Data and chart by National Collegiate Athletic Association. (n.d.).

NCAA Rules Timeline (Cole et al., 2026; Yurko, 2025)



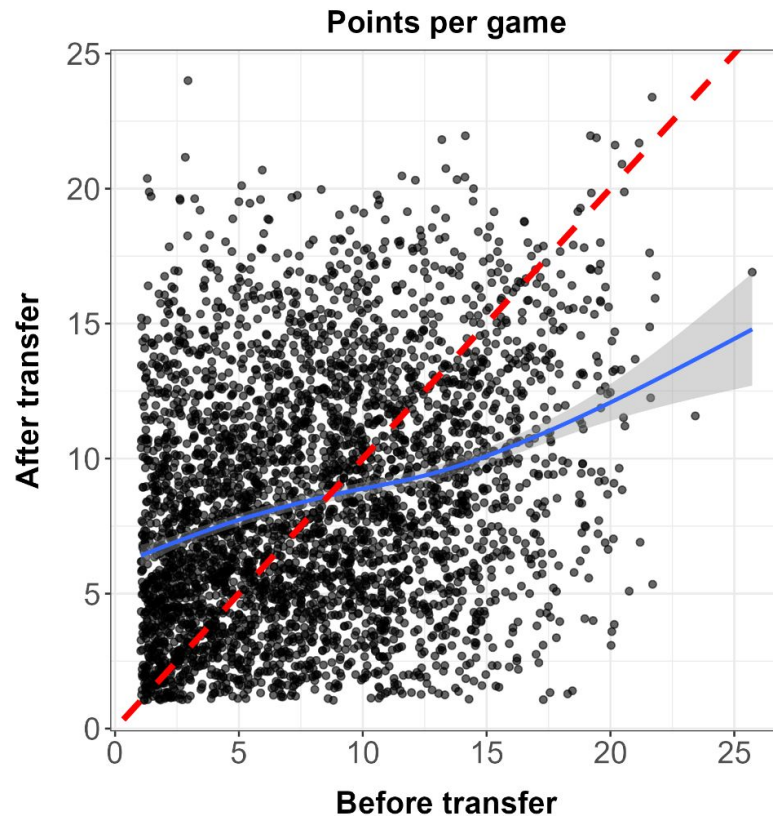
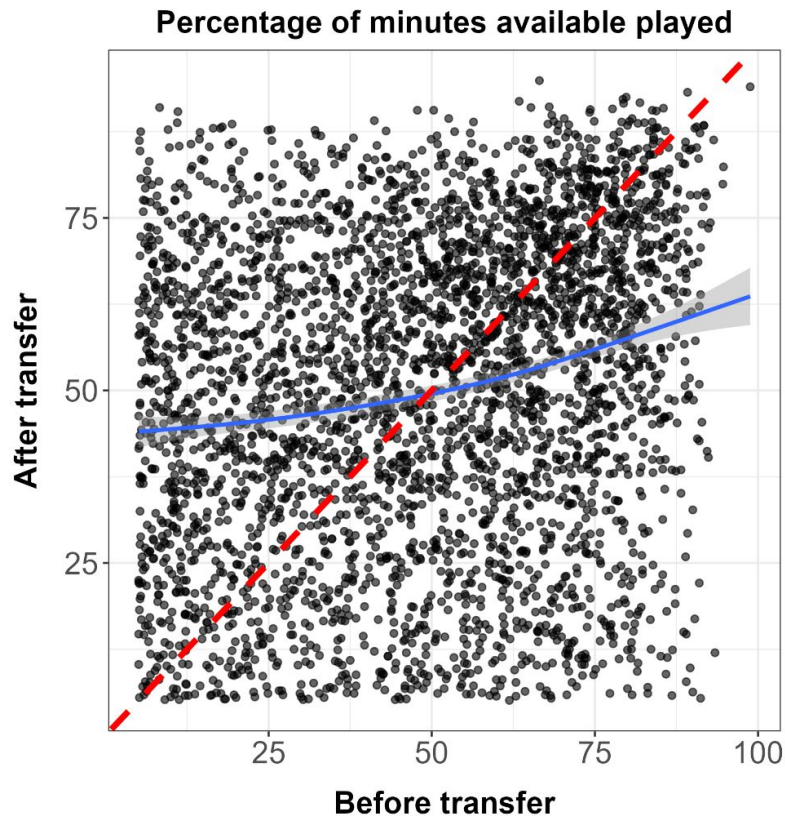
Data: hoopR package (Gilani, 2026) with additional preprocessing



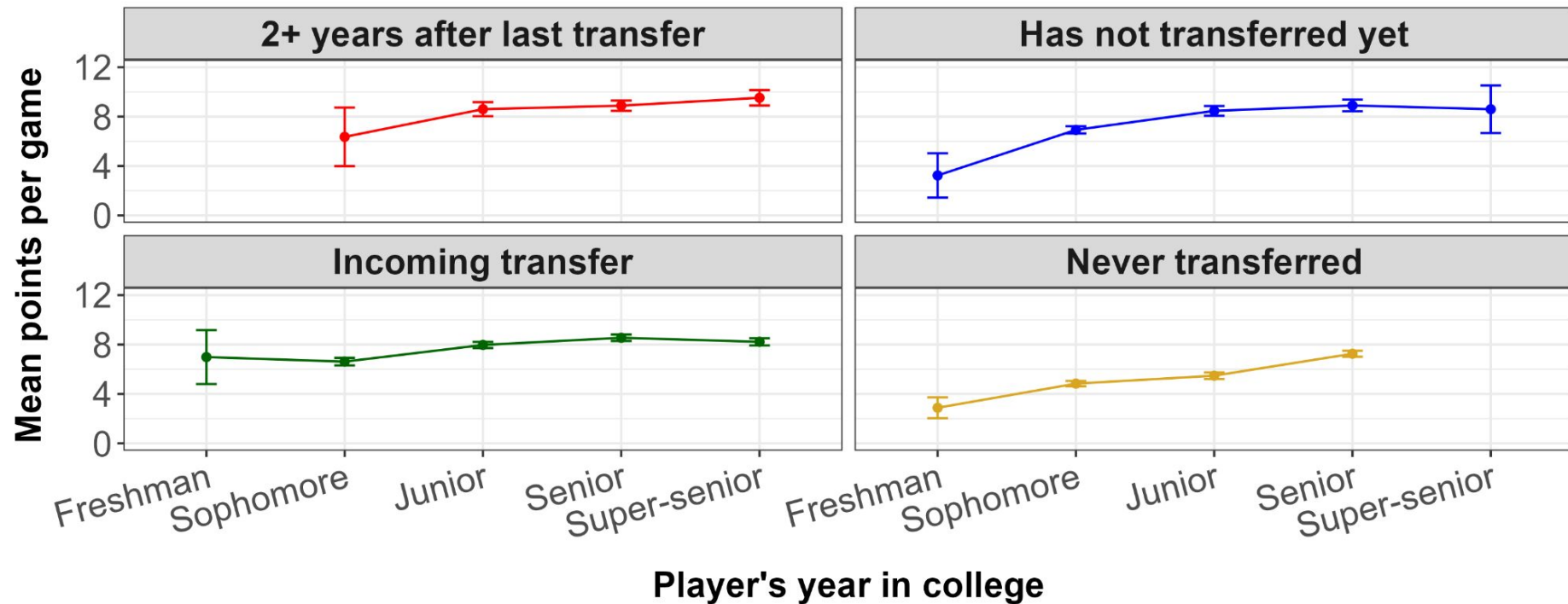
Player	Year	Team	PPG	Experience	Conference	Team Rating	Conference Rating	Transferred
H. Dickinson	2022	Michigan	18.6	Sophomore	B10	0.874	0.804	FALSE
H. Dickinson	2023	Michigan	17.8	Junior	B10	0.832	0.811	FALSE
H. Dickinson	2024	Kansas	17.9	Senior	B12	0.8976	0.838	TRUE
H. Dickinson	2025	Kansas	16.4	Super Senior	B12	0.921	0.855	FALSE

Data is retrieved from the Bart Torvik website (barttorvik.com). The data sample illustrates a particular player's development throughout their college basketball career.

EDA: Without controlling for any factors, if a player is already performing well, they should not transfer



EDA: Mean points per game generally improves over players' careers, but transfers typically outperform non-transfers



Error bars represent the corresponding 95% confidence interval.

Similar Players, One Transfers and One Stays

Otega Oweh

4 star guard @ Oklahoma

Transferred to Kentucky after Sophomore season



Sophomore Season

PPG = 11.4

MPG = 24.8



Junior Season

PPG = 16.2

MPG = 28.3

+4.8 Net PPG Change

Drafted 41st in NBA Draft

Chucky Hepburn

4 star guard @ Wisconsin

Stayed at Wisconsin for Sophomore and Junior season



Sophomore Season

PPG = 12.2

MPG = 32.0



Junior Season

PPG = 9.2

MPG = 33.3

-3.0 Net PPG Change

Went undrafted but later signed an NBA contract

Methods: Propensity score matching and mixed effects regression

- **Propensity score matching:** Matching each transferred player with a number of non-transferred players with similar statistics

$$\overline{\text{Transferred}}_{ijkt} = \beta_1 \text{Position}_i + \beta_2 \text{Experience}_{it} + \beta_3 \text{TeamRating}_{jt} + \beta_4 \text{ConfRating}_{kt}$$

- In the dataset: **Transferred** is a binary indicator variable (0 or 1) representing whether the player is an incoming transfer for that season

Notation: *i* represents a player, *j* represents a team, *k* represents a conference, and *t* represents a season (denoted by the year in which the NCAA championship was played).

Methods: Propensity score matching and mixed effects regression

- **Mixed effects regression:** Controlling for additional factors - player's development, team and conference context

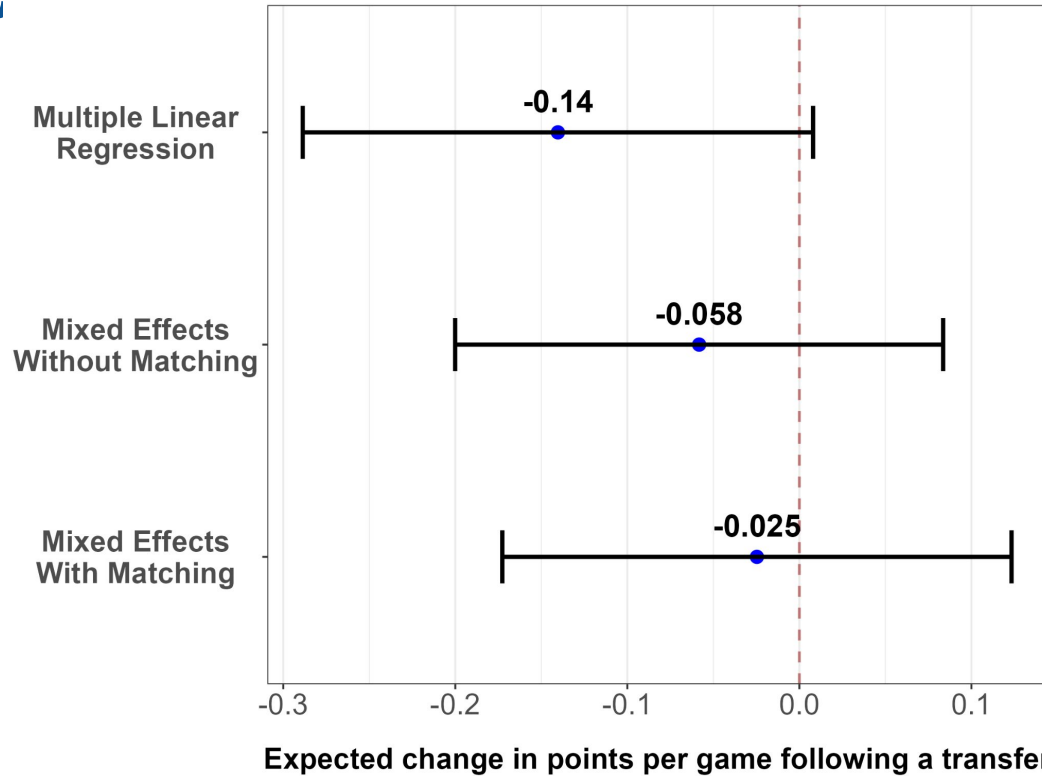
$$\widehat{\text{PPG}}_{ijkt} = a_{ij} + \beta_1 \text{Transferred}_{ij} + \beta_2 t + \beta_3 \text{PPG}_{i,t-1} + \beta_4 \text{Position}_i + \beta_5 \text{Experience}_{it} + \beta_6 \text{Conference}_{kt} + \beta_7 \text{TeamRating}_{jt} + \beta_8 \text{ConfRating}_{kt}$$

- Random intercepts for each player and team combination:

$$a_{ij} = \alpha_0 + u_{ij}, \quad u_{ij} \sim N(0, \tau^2).$$

Notation: i represents a player, j represents a team, k represents a conference, and t represents a season (denoted by the year in which the NCAA championship was played).

Transfers do **not** have a large nor significant impact on performance! ...or is it?



The numbers above each point represents the corresponding point estimate of the effect of a transfer on PPG. The error bars represent the 95% confidence intervals associated with each of the estimates.

Big Fish in a Little Pond, or Small Fish in a Big Pond?

Andrew Rhode



2023 - St. Thomas

PPG = 17.1
MPG = 33.0



2024 - Virginia

PPG = 4.3
MPG = 25.2

Alston Mason



2022 - Oklahoma

PPG = 1.4
MPG = 6.2

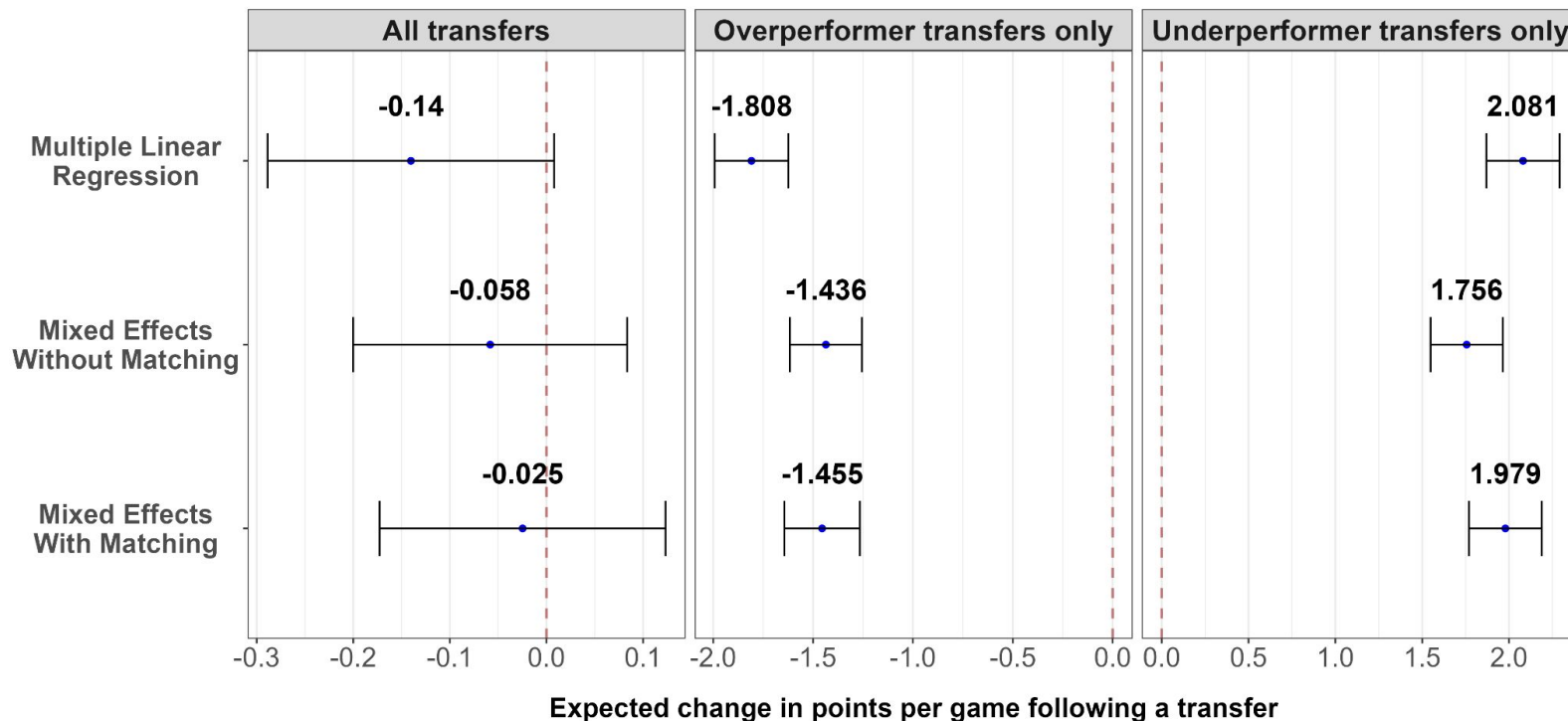


2023 - Missouri St

PPG = 9.9
MPG = 27.2

The “big fish, little pond” analogy (and its opposite) was inspired by a prior work on the subject by Cole et al. (2026).

Players who transferred as an overperformer tends to perform worse in the following season, and vice versa



The numbers above each point represents the corresponding point estimate of the effect of a transfer on PPG. The error bars represent the 95% confidence intervals associated with each of the estimates.

So what should the coaches do?

- Find underperformers at other programs in the transfer portal
- Thoughtfully analyze overperformers at different programs before extending an offer

And if I'm a player?

- If underperforming, don't pass up opportunities at a different program
- If finding success, reevaluate the decision of changing schools

Limitations & Next Steps

- The data does not fully capture:
 - Complex transfer histories: Redshirting, declaring for NBA drafts before end of eligibility, transfers involving non-DI institutions, etc.
 - Transferring for non-athletic reasons **and NIL**
- Points per game as indicator of player performance
 - Alternative metrics: Points per possession, RAPM, PORPAGATU!, etc.
- Model specifications
 - Heteroskedasticity in the matched data
 - Random slopes, GLMs and GAMs to capture non-linear relationships

References

- Cole, K., Pearman, F., & Khanna, S. (2026). Big Fish or Little Pond? National Evidence on the Relationship Between Transferring and Athletic Performance Within Division I Men's College Basketball. *Journal of Issues in Intercollegiate Athletics*, 19(1).
<https://doi.org/10.51221/sc.jiia.2026.19.1.15>
- Gilani, S. (2026). hoopR: Access Men's Basketball Play by Play Data. (Version 3.1.0) [Computer software]. <https://github.com/sportsdataverse/hoopR>
- National Collegiate Athletic Association. (n.d.). *Student-Athlete Transfer Research*. NCAA. Retrieved July 19, 2026, from
<https://www.ncaa.org/what-we-do/research/student-athlete-transfer-research/>
- Yurko, R. (2025, September 27). *College Football Volatility: A Bayesian State-Space Model of the Transfer Portal and NIL Impact* [Conference presentation]. 2025 New England Symposium on Statistics in Sports, Cambridge, MA, United States.
<https://www.youtube.com/watch?v=SdXpknMhGBc>

The Effect of Transferring on Player Performance:

An Analysis on NCAA Division I Men's Basketball:

1. Variable descriptions
2. Additional EDA plots
3. Love plots to assess balance in propensity score matching
4. Residuals versus fits plot for mixed effects regression model

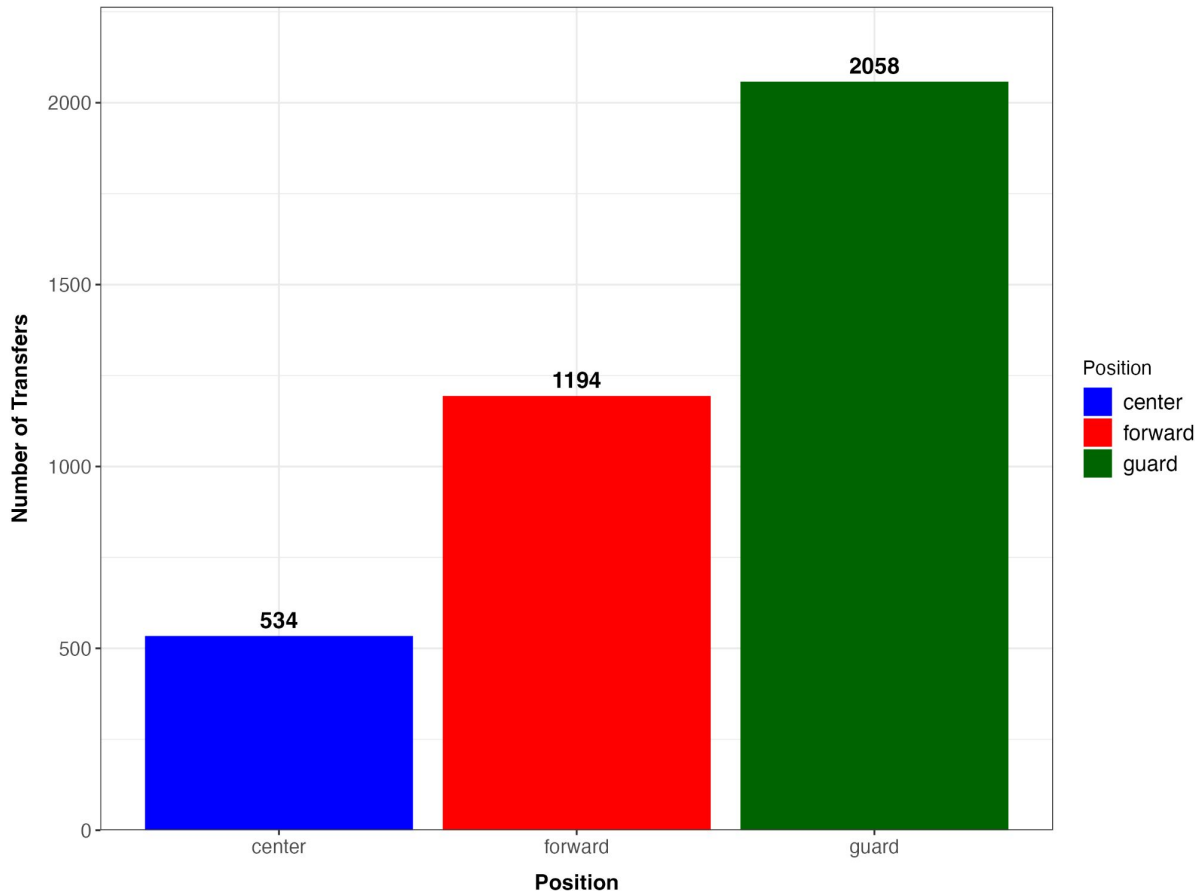
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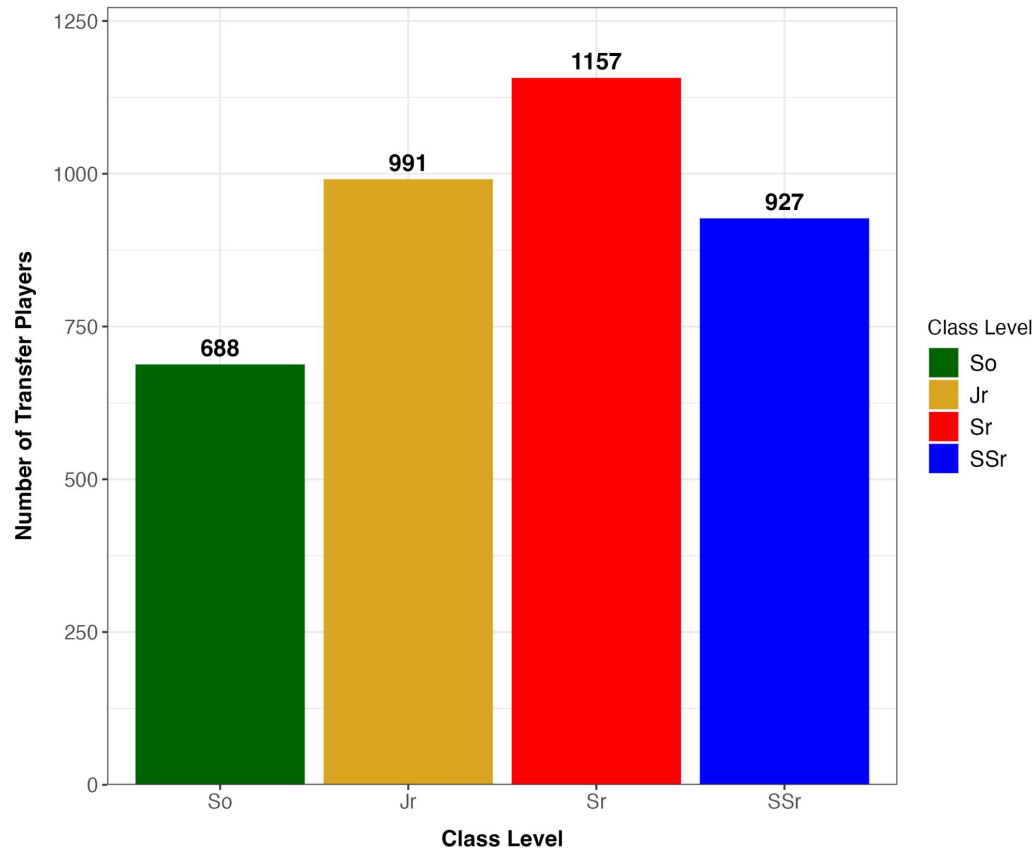
Appendix 1: Variable descriptions

Variable	Full name	Data type and additional preprocessing
PPG	Points per game	Numeric variable
Transferred	Is incoming transfer	Indicator variable: 1 if incoming transfer, 0 otherwise
Position	Position	Categorical variable: Center, Forward, or Guard
Experience	Experience level	Numeric variable: 1 (Freshman) to 5 (Super-seniors)
TeamRating	Team rating	Numeric variable: Barthag rating from Bart Torvik
ConfRating	Conference rating	Numeric variable: Average Barthag rating of all teams in a conference
Conference	Conference	Categorical variable

Appendix 2: Total number of transfers by broad position category

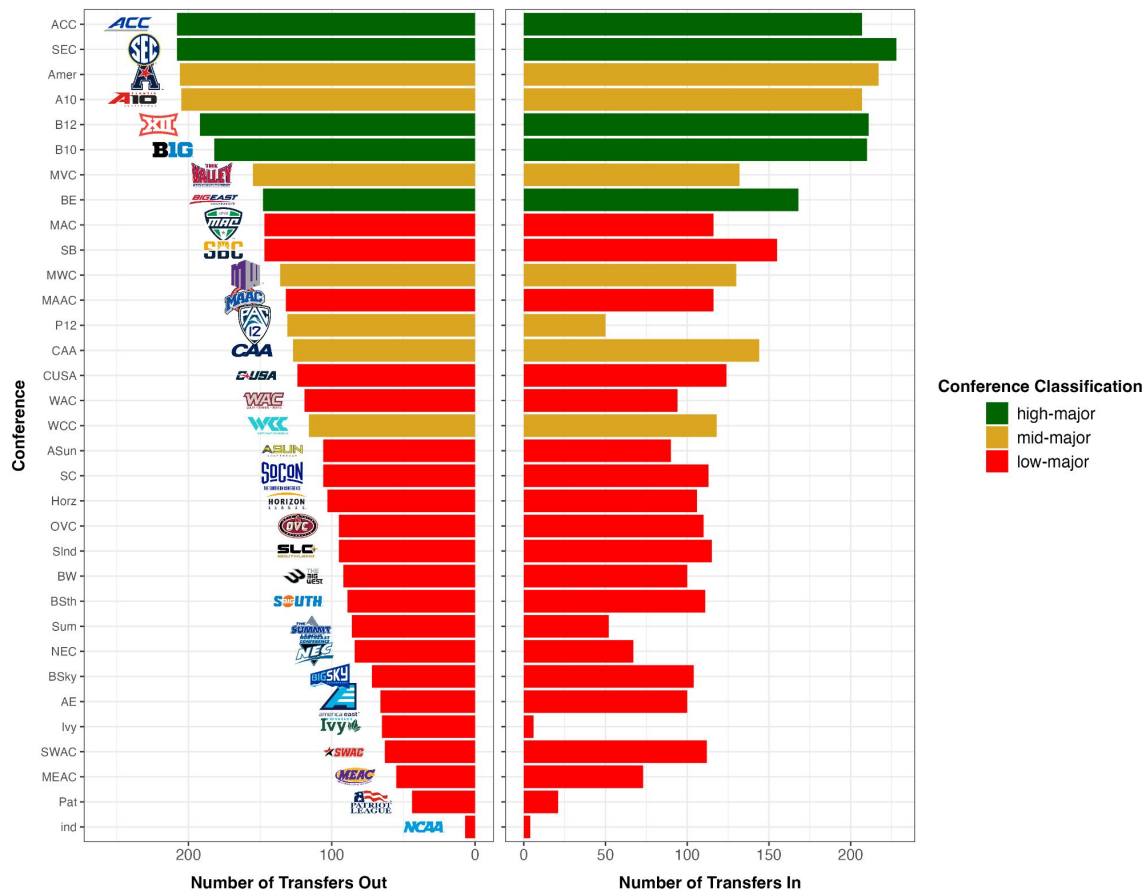


Appendix 2: Total number of transfers by class year



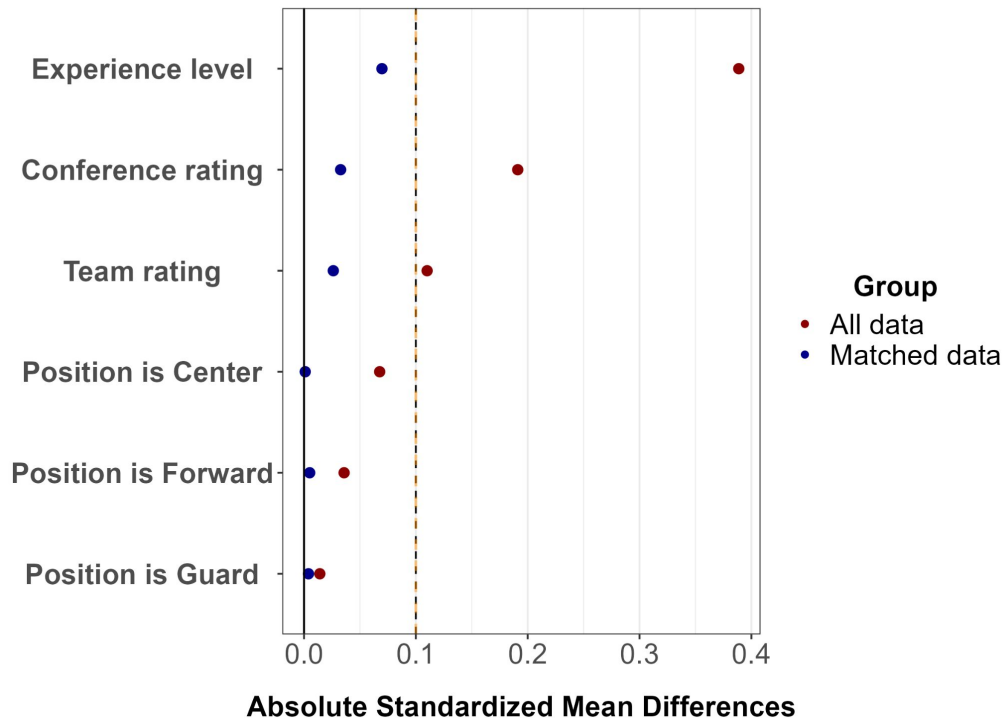
*SSr notation is short for Super Senior, or a player in their 5th or 6th season

Appendix 2: Total number of transfers in and out by conference



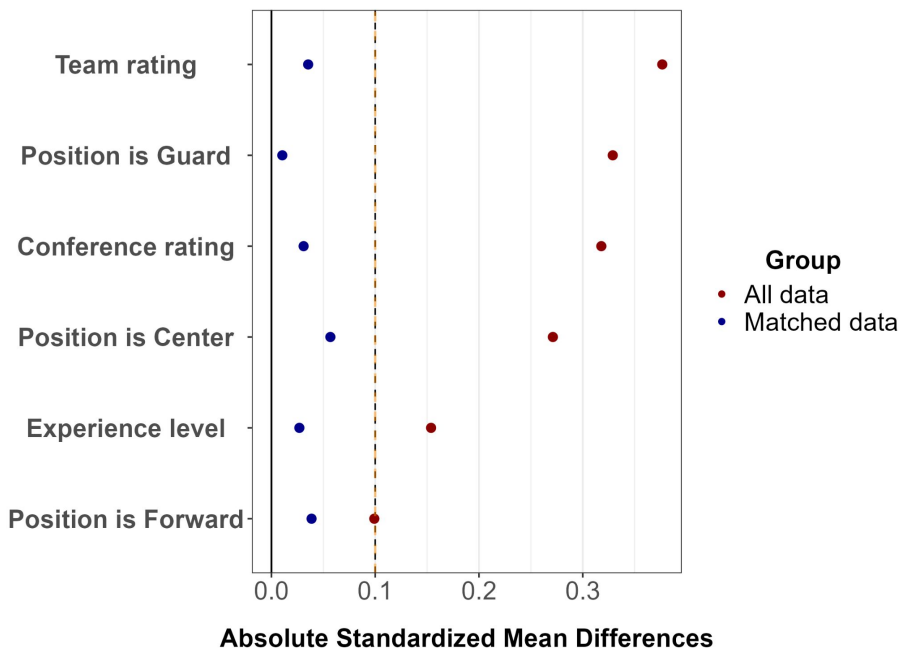
Appendix 3: Love plots to assess balance in propensity score matching

All transfers

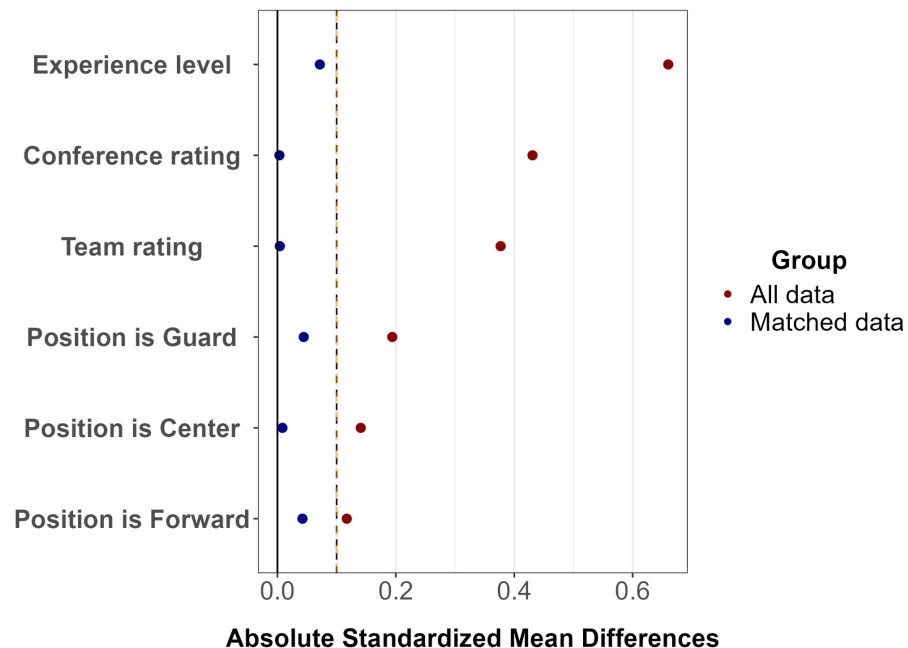


Appendix 3: Love plots to assess balance in propensity score matching

Underperformer transfers

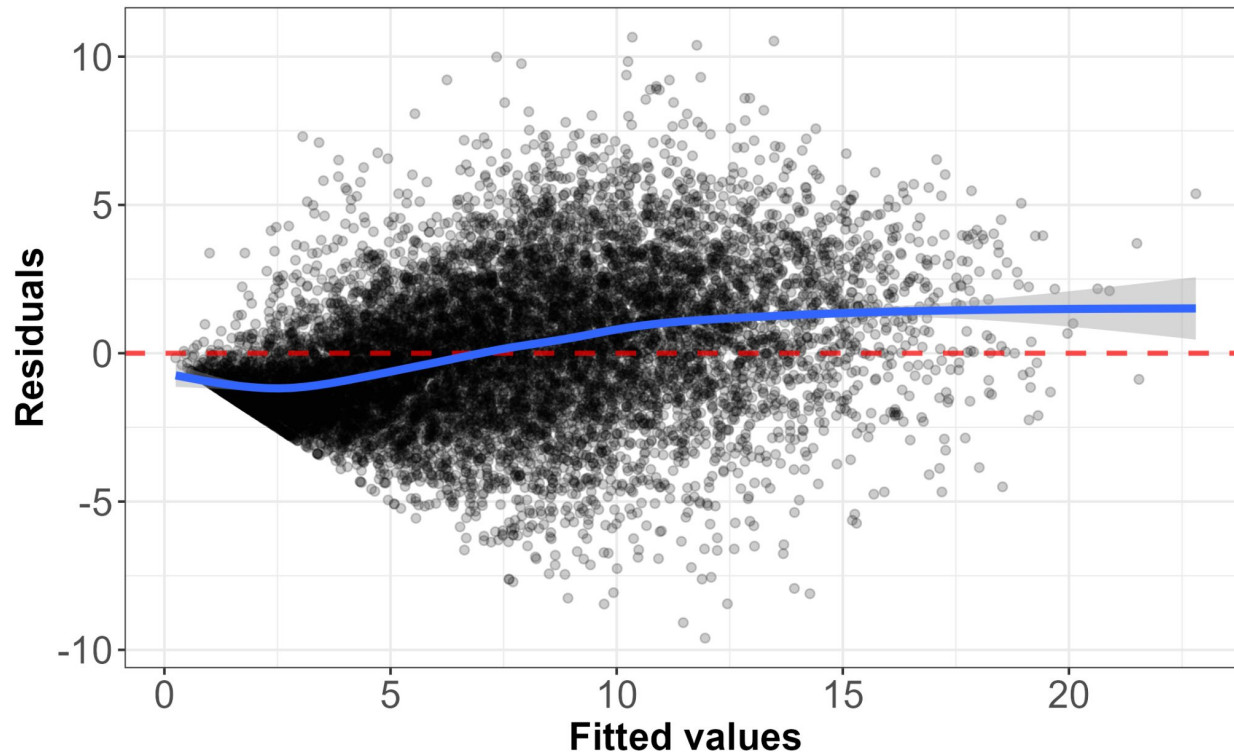


Overperformer transfers



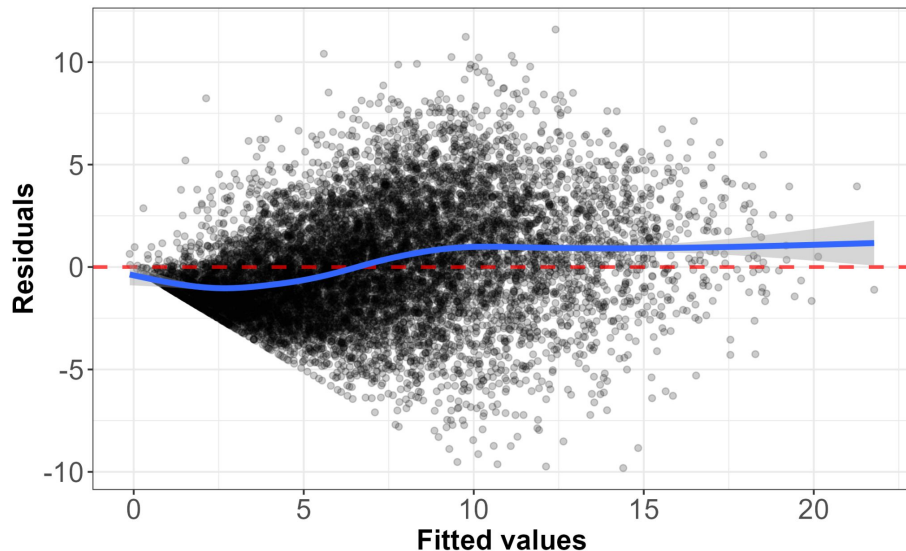
Appendix 4: Residuals vs. fits plot for mixed effects regression model

All transfers



Appendix 4: Residuals vs. fits plot for mixed effects regression model

Underperformer transfers



Overperformer transfers

